

The Geopolitics of Charcoal Production and Consumption in Nigeria: Energy or Climate?*

Odunola A. Oladeji¹

¹Bordeaux Montaigne University, France

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Abstract

With every passing United Nations Climate Change Conference without a concrete plan to phase out the use of coal in spite of the glaring climate change consequences, unanimously suggests continued reliance of nations on energy generated from coal and related products. While underdeveloped and developing economies might not contribute much within the context of usage, their counterpart developed economies possess a robust energy system that is largely or partly dependent on coal. Hence, failure to decarbonize the energy sector completely—inventing cheap alternative sources—would mean that complementary energy sources like charcoal will continue to be in demand. Apparently, the extractive nature of the economy obtainable in underdeveloped or developed nations continues to fuel the supply of charcoal even when it involves the destruction of forest biodiversity. This consequently constitutes a first-degree damage to the environment and secondly to the global climate. It is

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upon this background that this work seeks to study and analyze the geopolitics of charcoal production in Nigeria. Relying on existing literature, the methodological approach will be such that it critically examines the discourse within the purview of its socio-economic and environmental significance, as well as the potential trade-off.

Keywords: charcoal production, Nigeria, charcoal geopolitics, climate change, charcoal consumption

1 Introduction

Charcoal has, over the years, proven to be an indispensable source of energy considerably because of its domestic importance and commercial relevance. The cost of usage is such that the poor could afford, and the rich will pay any value to get. What that suggests is that charcoal is a significant energy source for both rural dwellers and urban metropolitans, both locally and globally (Ekpo and Mba, 2020). And because of this relative importance and interest, particularly in the international market, the demand for charcoal is on the rise, which in turn drives production in the source country - mostly developing countries (Adeniji *et al.*, 2015). More so, a major source of wood fuel is charcoal, often made from the controlled combustion of wood and wood materials such as logs, twigs, and any imaginable wood materials; could be the trunk or branches of trees or shrubs (Adeniji *et al.*, 2015; Zaku *et al.*, 2013). Once produced, charcoal serves as an energy source for use both domestically and industrially.

According to the Food and Agriculture Organization (FAO) estimate, the global consumption of charcoal is more than 40 million metric tonnes, while the domestic consumption fraction is estimated to be 2.4 billion of the global population that relies on charcoal and wood (Ekpo and Mba, 2020). Contextualizing this on a national scale, Nigeria is arguably

one of the largest producers of charcoal with a remarkable specialization in production across the Savannah belt of the country, from where it is distributed throughout the country and beyond. The main charcoal depots can be found in states like Oyo and Kwara in the western part of the country, Niger, Plateau and Kaduna towards the northern coordinate of the country (Adeniji *et al.*, 2015). This production capacity is second only to Brazil, but has a leading preference in the international market, considering the combustion effect that typical tropical hardwood charcoal gives—burning slowly and heats faster.

A combined effect of this sees Nigeria exporting a staggering 380,000 metric tonnes annually (Ekpo and Mba, 2020). Of course, this only accounts for the export share of the charcoal produced. On the consumption radar, Statistica reported a 75.4 percent share of energy derived from renewable sources in the primary energy consumption grid in 2018, although it is projected to increase to 86.4 percent by 2025 (Statistica, 2021). Complementing this, Ogunsawa and Ajala (2002) reported that the estimated daily consumption of firewood by rural communities in Nigeria is 27.5 million kg/day. From this sublime evidence, it is founded that fuel wood constitutes a greater proportion of this renewable energy. Moreover, amidst the quest for meeting basic energy needs for continued existence and survival, is the question of sustainability, which often situates the narrative within present exploitation and future availability. In another vein, critical to this discussion is also the challenging balance between the implications and impacts of continued charcoal production.

Markedly, the trade-off for the continued reliance on energy sources from wood fuel includes, among other things, loss of forest biodiversity and ecological functions as alluded to by Ekpo and Mba (2020). On second thought, the constant felling of trees for charcoal production also has a ripple effect on global climate. Meanwhile, the increasing commercialization of charcoal production driven by market demand both locally and globally has, without a doubt, created a network of economic opportunities, particularly for the rural economy in the economically less developed countries, as well as foreign exchange earn-

ings. At this point, the moral indictment in meeting energy needs through this source is conflated with the global injunction for climate best practices, which apparently advocate for afforestation as opposed to deforestation.

Hence, drawing on existing strands of literature, the contribution of this article is to contextualize the dynamics of charcoal production and consumption in Nigeria. Furthermore, the discourse will be situated critically along the recurrent debate of energy justice and climate change. To achieve this, the issue will be addressed from different fronts, and thus the remainder of this paper is organized to first of all appraise the socioeconomic significance. The following is a section on the biodiversity, environment, and sustainability cards. Afterwards, I examined the value chain of charcoal and, finally, the salient question of alternative energy use.

2 The socioeconomic significance of charcoal

Charcoal means different things to different people, and to imagine that the production and consumption will phase out soon is realistically non-concordant. Mapping this pragmatically, it will mean that the approximately 3 billion global population that depends on biomass—basically wood and charcoal—as a primary source of domestic energy will have to call it a quit (Salami and Brieger, 2011). Not only that, according to Adeniji *et al.*, (2015), 70 percent of households rely on wood fuel as their primary source of energy. Meaning, it will be inappropriate to overlook the role wood fuel, charcoal in particular, plays in the most basic energy requirement for sustenance. Not least, considering that forest resources account for 2.5 percent of gross domestic product in Nigeria (Adebayo *et al.*, 2019).

Despite having alternative energy sources, there still exists a large market demand for charcoal in developed nations such as the United States of America, Asia and Europe (Ekpo and Mba, 2020), at least for the love and prosperity of barbecue. For the largely

rural household in Africa, wood fuel, essentially firewood and charcoal, is both a means of survival and a source of livelihood. Explicably, it is the major energy source for cooking and heating, accounting for more than three-quarters of the primary energy supply in Sub-Saharan Africa (Bensch *et al.*, 2021). In Kenya, for instance, Ekpo and Mba (2020) reported that 82 percent of Urban household energy is provided by charcoal, as well as providing employment for over 700,000 people. The figure is higher (over 90 percent) in countries like Burkina Faso, Ethiopia, Mali, Tanzania and Zambia (Ibrahim *et al.*, 2021). According to the Food and Agriculture Organization report, in 2018, the value of charcoal market exceeded \$1.8 billion in annual revenue for 26 sub-Saharan African countries (Eniola, 2021).

In Nigeria, like its counterpart countries in Sub-Saharan Africa, both the largely rural households and urban dwellers rely on charcoal as their primary source of energy. Of course, it is cheap, at least compared to electric and kerosene stoves, gas or electric stoves (Rotowa *et al.*, 2019). And although Nigeria ranks first in Africa and second in the world in terms of production, the export ratio is meager; with Nigeria ranking 25th in the league of nations exporting charcoal (Rotowa *et al.*, 2019). Further corroborating this, according to Huitink (2018), Nigeria exported around 196 kilo tonnes of charcoal in 2016—that is about 5 percent of the initial production. The glaring inference from this is that Nigerians are the highest consumers of the charcoal produced in the country. In principle, both the production and the merchandise of charcoal flourish within its socioeconomic space, such that the export-retention mix is low because of high domestic consumption.

Furthermore, an insight from the analysis of Rotowa *et al.*, (2019) on the effect of indiscriminate charcoal production on the Nigeria forest estate revealed that the annual charcoal production in Nigeria in metric tonnes from 1990-2013 increased steadily year-on-year. From approximately 2.1 million metric tonnes in 1990 to 4.2 million metric tonnes in 2013, the supply steadily increased to meet the growing demand of the population. And ranging from domestic consumption to commercial use, charcoal serves as a source of energy

for cooking at the household level to outdoor events (Rotowa *et al.*, 2019), to serving as raw material for industrial production such as gunpowder, purifying/filtering agent in cottage industries and supply of heat for blacksmiths (Eniola, 2021).

Quite significantly, along the production-consumption matrix of charcoal, each gender is well represented and equally plays distinct roles. Several studies (Adebayo *et al.*, 2019; Chiteculo *et al.*, 2018; Adeniji *et al.*, 2015; Salami and Brieger, 2011) have established that males dominate the production segment. Whereas Salami and Brieger (2011) highlighted the intricate role women play in the management of forests through the routine cutting and subsidizing of firework needs. The illustration of the charcoal value chain in Bie province of Angola by (Chiteculo *et al.*, 2018) revealed that men dominated the production and transportation segment of the value chain, while the part of the trading and consumption was largely driven by women, including the sales of bags used for packaging. In commensurate terms, this also coincides with the findings of the existing literature on charcoal production in Nigeria. In addition to this, Salami and Brieger (2011) scholarship on commercial charcoal production in the Ibarapa district of Southwestern Nigeria highlighted how the charcoal industry is creating jobs for youths throughout the charcoal's value chain. Inadvertently, from this evidence, charcoal not only serves as a source of energy for cooking, but also offers a wide array of opportunities with tangible economic significance for both rural households and urban dwellers.

3 From Forest to Fire: The Charcoal Production Value Chain

The charcoal value chain is enormous, involving a series of linked operations and chains from production to consumption. In particular, this continuous chain also involves the different stakeholders operating at each specified level (Salami and Brieger, 2011). However, what ended up being a clean cooking energy source in the different households, whether in rural or urban regions, was originating in the woods. Rotowa *et al.*, (2019) and Mba

(2018) both agreed that charcoal production is initiated first in rural areas where forests and forest reserves are found, and that wood is in abundance for production. More importantly, the presence of tropical woods in Nigerian forests lists the country as one of the few countries that produces wood charcoal and the distribution is spread throughout all geopolitical zones of the country. However, the main depots are located in the Southwest, Northwest, and North Central parts of the country (Rotowa *et al.*, 2019). From these major depots, charcoal is distributed in large numbers to urban centers and major ports for export.

Essentially, from forest to fire, the value chain analysis for charcoal spans from the procurement of raw material, that is, biomass (wood), to the carbonization process, then packing, bagging, transportation, distribution and finally consumption (United Nations Development Programm, n.d.). In the production stage, charcoal producers source trees, fell them, construct the traditional kiln system, and then start the burning and opening process (Mba, 2018). Although all wood species can be carbonized to charcoal, not all species provide the quality of charcoal desired for consumption. This particularly forms an underlying basis for understanding the preferences of producers. From Adebayo *et al.*, (2019) finding on producers' choice of trees used for charcoal production in the producing region of Oyo state, the choicest species include Shea tree (*Vitellaria paradoxa*) and Axle wood (*Anogeissus leiocarpus*), followed by Guinea wood (*Bridelia ferruginea*), Tropical almond tree (*Terminilia spp*), and Pericopsis (*Pericopsis laxiflora*).

A similar study by Ekpo, Ayuba, *et al.*, (2021) on the analysis of tree species preference among commercial charcoal producers in Nasarawa state discovered around 22 trees species for charcoal production in the study area. Like its counterpart producing region in the Southwest, the Shea tree (*Vitellaria paradoxa*), the Axle wood (*Anogeissus leiocarpus*), the Tropical almond tree (*Terminilia spp*), and the Pericopsis (*Pericopsis laxiflora*) are in the ranks of preferred species that are also commonly used in Nasarawa, which is in the North central part of the country (Ekpo, Ayuba, *et al.*, 2021). Once production is complete, pack-

ing and bagging are another stage of engagement along the charcoal value chain. This alone provides employment opportunities for rural dwellers. Inextricably linked to this stage are the suppliers of bagging materials who also make a living from the sales of bags. Other closely related operations at this stage are logistics and transport services.

Consequently, charcoal is loaded and ferried in containers, especially commercial quantities that are export-bound or transported via the commonly used trucks or lorries. In quantifying charcoal commuting, Ekpo, Ayuba, *et al.*, (2021) noted that the Nigerian government reported an estimated 150 trucks transporting charcoal to markets on a weekly basis from the North Central alone. However, they could be stored or sent to the depots. Regarding ease of storage, Eniola (2021) contended that, unlike firewood, charcoal uses a small space, is easy to handle. and does not easily deteriorate. In addition, the overall functioning of the value chain provides employment opportunities for the teeming rural populace, as well as urban dwellers. Hence, the contribution of charcoal production to the socioeconomic functioning of the rural economy cannot be underestimated.

4 Biodiversity, Environment and the Sustainability Card

The constellation of factors that lead to the production of charcoal is a testament to the age-long interaction and dependence of humans on the environment. These anthropogenic activities of humans can contribute to or be detrimental to both the biological and physical environments. Also, it should be noted that the degree of dependency on this environment varies from country to country. For example, in developing countries where about 70 percent of the population live in rural communities and depend greatly on their immediate environment for their survival, the burden on the available natural resources is not the same as in developed countries (Rotowa *et al.*, 2019). According to Carney (1998), the global reliance on biomass fuel for cooking and heating is responsible for the astronomical rise in the world's charcoal production between 1970 and 1995. Consequently, as

reported by Oriola and Omofoyewa (2013), there is approximately 50 percent deforestation of global forest areas.

Arguably, charcoal production constitutes a large part of this forest depletion. In 25 years, precisely from 1990 to 2005, the Food and Agriculture Organization revealed that Nigeria lost 35.7 percent of its forest cover (Rotowa *et al.*, 2019). In 2005 alone, the Forest Resources Assessment Country report showed that the total wood removals from Nigerian forests were 86,626,797 m³, of which the removals for wood fuel from forests were 72,710,935 m³ (Ogundele *et al.*, 2011). This uncontrolled felling of trees for charcoal production not only leads to the depletion of forest resources, but also poses a threat to forest biodiversity. Oyo State, situated in the southwestern part of Nigeria and one of the most prominent hubs for charcoal production, decried the gradual replacement of the initial primary forest land with secondary growth due to intensive cultivation and overexploitation. Compared to the past, when the state could boast of abundant forest reserves, today it could only account for nine forest reserves (Salami and Brieger, 2011). With the indiscriminate production of charcoal in different production centers and the consequent loss of forest biodiversity, the Nigerian government placed a ban on charcoal exportation because continuous charcoal production failed to meet the tree replacement policy (Oritse, 2016).

Besides the depletion of forest resources, other intrinsic components of charcoal production include soil and climatic impacts, since all of these constitute the nature and structure of forest biodiversity (Salami and Brieger, 2011). According to them, the processes that lead to the production of charcoal affect the structure of the soil; the structure of the soil is altered as a result of the pit created during the construction of the kiln. Similarly, heat is released during the combustion of wood that gives rise to charcoal, leading to the vaporization of certain nutrients from the soil. Additionally, studies (Ogundele *et al.*, 2011; Oguntunde *et al.*, 2008) have shown that the soil around the charcoal production site is slightly higher in pH, has a characteristic bulk density of surface temperature and albedo, higher infiltration rate, among others (Eniola, 2021). Likewise, continuous deforestation

exposes the soil to erosion, climate change, and threats to forest diversity. Unlike the emission of charcoal burning, which is relatively mild, the emission during charcoal production is quite significant (Eniola, 2021). According to Bailis *et al.*, (2015), on the continent of Africa, Nigeria tops the list of countries with the highest absolute emissions from charcoal production.

Without controversy, the energy from wood fuel, particularly charcoal, has served the majority of households in Sub-Saharan Africa. However, charcoal production is largely unsustainable, and certain species of wood that were abundant in recent years are gradually becoming extinct. Then it becomes imperative to address holistically the energy needs of the continent, sustainable wood fuel production and usage, and the preservation of forest biodiversity.

5 Where are the alternatives?

Incontestably, a huge dependence on wood fuel as energy comes at a cost to the environment and the climate at large. The indiscriminate felling of wood for charcoal, as well as the increase in the rate of deforestation in the producing regions, are reliable indicators of this assertion. For developed economies, it is expected that they have ascended the fuel ladder, which according to Zaku *et al.*, (2013) is a construction attributed to the transitions from wood fuel in the urbanized area due to an increase in income level. Hypothetically, this decline in the wood fuel mix as countries transit from developing to developed economies, as argued by Zaku *et al.*, (2013), is true in Asia, where the consumption of fuel wood was found to have been replaced with alternative fuels as the continent experienced rapid economic growth.

On the contrary, the continent of Africa, which has yet to evolve to its full potential as far as development is concerned, still largely relies on energy from wood fuel. According to

Salami and Brieger (2011), 56 percent of the total energy consumed in Africa is derived from wood and charcoal. And, as reported by Harvard School of Public Health, in 2000, homes in Sub-Saharan Africa consumed around 470 million tonnes of wood. Except for South Africa and countries at the extreme end of North Africa, the rest of Africa relies a great deal on wood fuel, the share ranging from 61 percent to 86 percent of primary energy consumption (Amous, 2005).

In Nigeria, the energy mix is largely driven by fossil fuels and wood fuels in spite of the several other endowments such as natural gas, wind, solar, and biomass (Eniola, 2021). In fact, reports revealed that Nigeria ranks as the seventh largest country with gas reserves, with an estimated reserve of about 186 trillion cubic feet (Rotowa *et al.*, 2019). Yet, the share of primary energy sources in the country showed that 10.4 percent of the energy supply is from gas, hydro contributed a meager 0.6 percent and commercial renewable energy constituted the highest percentage – 83 percent. Of which, wood and other agricultural waste constitute the commercial renewable energy (Zaku *et al.*, 2013). How these huge potentials for energy generation have not been translated to diversified energy sources other than the traditional wood fuel–charcoal in particular—is a concern worth examining.

Inarguably, charcoal remains a cheap, if not the cheapest energy source. And this explains the economics of preference that plays out especially in most developing economies with a largely rural populace (Eniola, 2021). Beyond just the consumption of charcoal, the attachment and dependence on charcoal are significant because it contributes more and also comes with many opportunities for people. Take, for example, the market value of charcoal in 26 countries in Sub-Saharan Africa alone, as reported by the Food and Agriculture Organization, reportedly exceeds 1.8 billion dollars yearly (Eniola, 2021). In addition, charcoal provides a means of livelihood for several rural households, as well as significant employment opportunities for both rural (at the production and off-take level) and urban dwellers – that is, suppliers and retailers.

From Shackleton *et al.*, (2006) findings, wood fuel and charcoal production have become a source of income for about 65 percent of rural dwellers. However, when inquiring into the dynamics of heavy dependence on wood fuel and charcoal but limited dependence on alternative energy, particularly in Nigeria and other countries in Sub-Saharan Africa, there are a constellation of factors such as poverty incidence, unemployment rate, etc. (Eniola, 2021; Zaku *et al.*, 2013). For example, the poverty rate and percentage of wood fuel by geopolitical regions released by the National Bureau of Statistics in 2017 showed a corresponding dependence on wood fuel as the poverty rate increases. The North-eastern part of the country has an estimated 72.2 percent poverty rate and a 95.9 percent dependence on wood as a fuel source. Following closely is North-west with a 71.2 per cent poverty rate and a 95.3 per cent dependence on wood as a source of fuel. Only the South-west seems to be the odd one out of this progression – about 43 percent poverty rate and the lowest (54.9 percent) dependence on wood as fuel despite having abundant tropical wood for charcoal production. This could be due to increased urbanization in the region, proximity to alternative sources such as gas, and other factors not listed (Zaku *et al.*, 2013).

The unemployment rate in the country is also a potential trigger for rural dwellers venturing into mass charcoal production. Similarly, the almost non-existent basic amenities in rural areas, the inaccessibility of alternative energy sources such as electricity, cooking gas, and solar energy (which is even more expensive despite the abundant sunshine in the tropics) in rural corridors is a substantive causation for massive production and consumption to meet their basic energy needs (Eniola, 2021). In some other cases, the privatization of this alternative – electricity, for example – makes it unaffordable to largely low-income households, making charcoal a better and more affordable alternative (Salami and Brieger, 2011). Also, in recent times, the cost of agricultural inputs (seeds, fertilizers, etc.) has become exorbitant, leaving farmers with no other alternative than relying on charcoal production as a source of livelihood (Eniola, 2021).

However, for most rural dwellers involved in the production of charcoal, the implication

of their action on the environment is not a priority because the instinct to survive far outweighs the consciousness of the implications on the environment. Hence, the unsustainable production of charcoal is likely not stopping anytime soon, which is a serious challenge to the environment and a catalyst for the impact of climate change. The loss of forest biodiversity will continue to linger whilst felling of forest woods will continue to become the order of the day, and what happens when the entire forest reserve and tree cover is depleted? This impending damage could be mitigated by enacting stringent policies on sustainable charcoal production, as a total ban seems infeasible. Similarly, untapped alternative energy sources need to be developed to full capacity. That will cross the benchmark on the availability and affordability of energy for the largely poor households.

6 Conclusion

The evidence gathered from the study revealed that charcoal is a major source of energy for cooking and heating in less developed economies. In addition, it is a significant economic driver in rural areas in addition to agricultural production. For Nigeria, the charcoal market is massive and a significant proportion of charcoal production is consumed by its citizens, both rural households and urban dwellers. The major depots are spread across the Southwest-, Northeast-, and North Central geopolitical zones. In particular, only a paltry portion of what is produced is sold to the export market.

Considering that charcoal is a cheaper energy source compared to the other available alternatives in the country, the demand as well as the supply of charcoal have been on the rise over the years. The available alternative sources, such as fossil fuels or gas, are either expensive or largely untapped. Meanwhile, findings have revealed that the incidence of poverty in the country, high unemployment, among others, are broader issues surrounding charcoal production. Without controversy, the cost of this massive production and consumption of charcoal is on the environment; the loss of forest diversity, the depletion

of forest reserves, and the extinction of certain wood species due to uncontrolled felling of trees.

In a deeper analysis, the combined effect of all these contributes to climate change. But for charcoal producers and consumers in developing countries such as Nigeria, where alternatives are almost non-existent because they are either expensive or unavailable to the larger populace, the moral indictment to meet the energy need for survival conflates with the global concern for climate change. Hence, a more plausible solution is to promote sustainable charcoal production in Nigeria and, by extension, Africa. Otherwise, transitioning up the energy ladder away from wood fuel means fully exploring and developing alternative sources that are cheaper and more accessible to both rural households and urban dwellers.

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